

Name: _____

Date: _____

VECTORS,

(ADDING AND SUBTRACTING VECTORS)

LO: I can add and subtract column vectors and represent the results geometrically.

Vector addition and subtraction

A **vector** has both magnitude and direction. **Column vectors** are written as (x, y) where x is horizontal and y is vertical movement.

Rules: To add vectors, add corresponding components. To subtract, subtract components. $a + b$ means 'go along a then along b '.

Quick examples

●	$(2,3) + (4,1) = (6,4)$	add components
●	$(5,2) - (3,1) = (2,1)$	subtract components
●	$-a$ is vector a reversed	negate both parts
●	$(2,3) + (4,1) = (8,3)$	add each part separately
●	Vectors can only be positive	can have negative components

Key idea

Add or subtract the x -components and y -components separately.

$$(a,b) + (c,d) = (a+c, b+d)$$

Component-wise addition

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - adding vectors

Find $(3,2) + (4,-1)$.

Add x : $3 + 4 = 7$

Add y : $2 + (-1) = 1$

Result: $(7, 1)$

$(7, 1)$

Add each component separately.

Example 2 - subtracting vectors

Find $(5,3) - (2,7)$.

Subtract x : $5 - 2 = 3$

Subtract y : $3 - 7 = -4$

Result: $(3, -4)$

$(3, -4)$

Subtract each component.

Example 3 - negative vector

If $a = (4, -2)$, find $-a$.

Negate x : -4

Negate y : 2

Result: $(-4, 2)$

$(-4, 2)$

Reverse both components.

Example 4 - resultant vector

Go from A to B using $a = (3,1)$ then $b = (2,4)$. Find the resultant.

Resultant = $a + b$

= $(3+2, 1+4)$

= $(5, 5)$

$(5, 5)$

The resultant goes directly from start to finish.

YOUR TURN – PRACTICE (deliberate practice)

1. Adding vectors

Find $a + b$.

- a) $a=(2,5)$, $b=(3,1)$
- b) $a=(4,-2)$, $b=(1,6)$
- c) $a=(-3,2)$, $b=(5,-4)$
- d) $a=(7,0)$, $b=(-3,4)$

2. Subtracting vectors

Find $a - b$.

- a) $a=(6,3)$, $b=(2,1)$
- b) $a=(4,5)$, $b=(7,2)$
- c) $a=(1,-3)$, $b=(-2,4)$
- d) $a=(0,8)$, $b=(5,-1)$

3. Negative vectors

Find the negative of each vector.

- a) $(3, 7)$
- b) $(-2, 5)$
- c) $(4, -6)$
- d) $(-1, -8)$

4. Combined operations

Calculate each expression.

- a) $2a + b$ where $a=(1,3)$, $b=(4,-2)$
- b) $a - 2b$ where $a=(5,1)$, $b=(2,3)$
- c) $a + b + c$ where $a=(1,2)$, $b=(3,-1)$, $c=(-2,4)$
- d) $3a - b$ where $a=(2,1)$, $b=(4,-3)$

5. Mixed vector problems

Solve each problem.

- a) $AB = (3,4)$. Find BA .
- b) $AB=(2,5)$, $BC=(3,-1)$. Find AC .
- c) $a+b = (7,3)$, $a = (4,1)$. Find b .
- d) If $OA=(3,2)$ and $OB=(7,5)$, find AB .
- e) Midpoint of AB where $A=(2,4)$, $B=(8,10)$. Position vector of M ?

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) To add vectors, add corresponding components ☐

Correct answer: _____

Reason: _____

b) $(2,3) + (4,1) = (6,3)$ ☐

Correct answer: _____

Reason: _____

c) $AB + BA = 0$ (zero vector) ☐

Correct answer: _____

Reason: _____

d) Vectors must have positive components ☐

Correct answer: _____

Reason: _____

e) The resultant vector gives the direct path ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) ABCD is a parallelogram. $AB = (4, 2)$ and $AD = (1, 5)$. Find the vectors AC and BD . Show that the diagonals bisect each other.

Expression: _____

Simplified: _____

b) Three forces act on an object: $F_1 = (3, -2)$, $F_2 = (-1, 5)$, $F_3 = (-4, -1)$. Find the resultant force. What fourth force would balance the system?

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

